Last Week

1. [**Literature**]: Consult the SOTA for AGReIDv2 and the protocols from BAU, and benchmark the current top Finsler model against them. Decide or not to pursue publication given results.
2. [**Training**]: Train the baseline and best model configuration with full 60 epochs
3. [**Training**]: Train the best model configuration on one of BAU's training dataset
4. [**Distance**]: Use slerp interpolation to integrate on a spherical geodesic for option 2
   - Consider methods to learn omega on the tangent plane and enforce that
   - Example: Identity – XX^t is on the tangent of the sphere, where x is normalized
5. [**Model**]: Enforcing tangent plane orthogonality
6. [**Model**]: Implement heuristic regularization terms to prevent divergence of the omega norm
7. [**Theory**]: Explain why drift align and bidirectional triplet are incompatible regularization terms together
8. [**Theory**]: Explain why having a 32-dimensional drift vector is optimal, compared to 64 and 16; and why the drift vector does not diverge to the max in this specific scenario

[**Sanity Test**]: Devise a small experiment to test the logic behind cross-res asymmetry and camera-viewpoint bias (e.g., small dataset + simple metric).

[**Literature**]: Current SOTA for AGReIDv2 and BAU

Found that the premise of integrating an asymmetric Finsler space into a domain-generalizable person Re-ID space is theoretically challenging

- Person Re-ID is fundamentally an identity retrieval task, which necessitates a equivalence relation governed by metric symmetry
- Introducing distance asymmetry inherently violates identity equivariance, as the retrieval ranking becomes strictly dependent on the query-gallery orientation.

Train: Market1501 + MSMT17 + CUHK-SYSU → Test: CUHK03

- **CLIP-ReID with Depth-First Graph Sampling:** 50.4% mAP, 51.1% Rank-1
  - Architecture: ViT-B/16 with text-token prompt learning and graph-based hard negative mining.
- **BAU:** 42.8% mAP, 43.9% Rank-1

AG-ReIDv2: Aerial → Camera

- **V2E (Baseline):** 80.72% mAP, 88.77% Rank-1.
  - Architecture: Three-stream ResNet/ViT hybrid with explainable elevated-view attention.
- **SD-ReID**: 80.61% mAP, 87.04% Rank-1.
  - Generatesview-specific features via stable diffusion to bridge extreme perspective gap

The equality operator over a set of identities constitutes a mathematical equivalence relation because it inherently satisfies three axioms: reflexivity, symmetry ,and transitivity

Because the underlying identity relation is symmetric ($y_i = y_j$ is identical to $y_j = y_i$,(the continuous distance function used to measure their similarity must also be symmetric:

$$d(x_i, x_j) = d(x_j, x_i)$$

.If a distance function is asymmetric (as in a Finsler/Randers metric where $F(x_i, x_j) \neq F(x_j, x_i)$,(it violates the geometry of the equivalence class. It implies

$$x_i$$

is closer to

$$x_j$$

than

$$x_j$$

is to

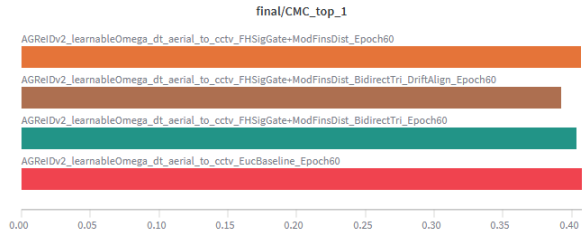
$$x_i$$

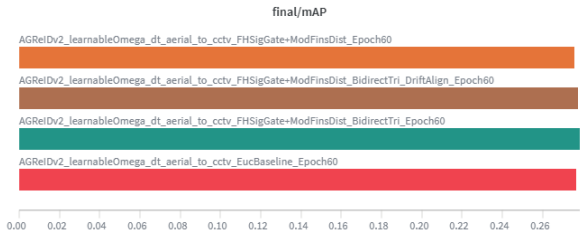
,which contradicts the binary, bidirectional nature of identity matching.

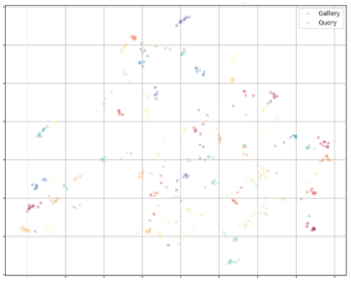
- **Incompatible Objectives:** BAU is a DG-ReID framework. Its primary objective (alignment and uniformity) is to suppress domain-specific stylistic variations (e.g., lighting, resolution) to extract a universal, domain-agnostic identity representation.
- **Deterministic Geometry vs. Style Shift:** AG-ReID.v2 exhibits severe deterministic geometric transformations. This is not a "style shift." SOTA approaches on this dataset (like SD-ReID and V2E) explicitly retain and model the view-specific structure rather than projecting it into a uniform hypersphere.
- **Topological Conflict:** Grafting an asymmetric Finslerian drift vector ω onto BAU and evaluating it on AG-ReID.v2 means your loss function is simultaneously trying to collapse domains into a uniform hypersphere (via BAU's uniformity loss) while trying to force directed, asymmetric distances based on structural elevation (via the Randers metric). These gradients will destructively interfere.

- If your objective is to evaluate a Finslerian manifold for asymmetric cross-view matching, you must benchmark against AG-ReID.v2 SOTAs using view-aware architectures, completely abandoning BAU and its uniformity constraints.
- If your objective is domain generalization, you must evaluate on the M+MS+CS → C3 split using standard VFM baselines like CLIP-ReID, and justify mathematically how an asymmetric distance metric preserves cross-domain identity equivariance.

Training on AGReIDv2 for 60 Epochs







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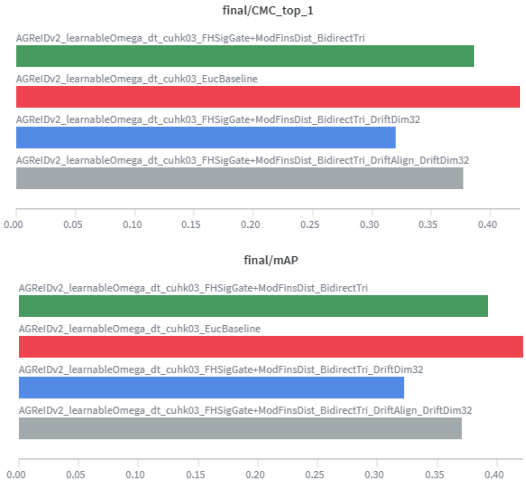
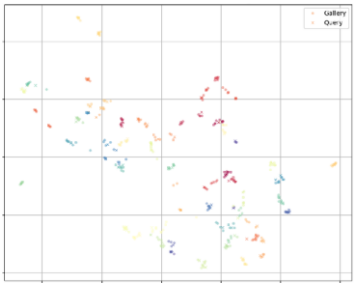
Baseline

Training on M+MS+CS → C3 (Protocol 2) for 60 Epochs





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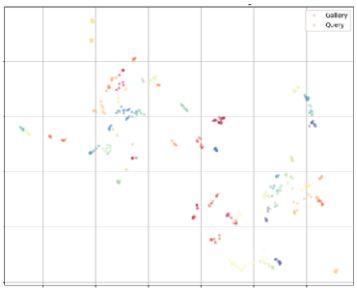


Baseline

[Distance 1] Using Slerp Interpolation

- **Spherical geodesic as path**: Taking spherical geodesic as the path between x and y
- **Field Assumption**: The drift vector field w changes linearly along this path

1 $$\mathcal{A}(\mathbf{x}, \mathbf{y}) = \int_0^1 \langle \omega(\gamma(t)), \dot{\gamma}(t) \rangle dt$$

2 $$\theta = \arccos(\langle x_{id}, y_{id} \rangle) \quad | \quad \gamma(t) = \frac{\sin((1-t)\theta)}{\sin(\theta)} x_{id} + \frac{\sin(t\theta)}{\sin(\theta)} y_{id}\,, \quad t \in [0, 1]$$

3 $$\dot{\gamma}(t) = A(t) x_{id} + B(t) y_{id}$$

where, $$A(t) = -\frac{\theta}{\sin(\theta)} \cos((1-t)\theta) \quad , \quad B(t) = \frac{\theta}{\sin(\theta)} \cos(t\theta)$$

4 $$\omega(t) \approx (1-t) x_\omega + t y_\omega$$
$$\langle (1-t) x_\omega + t y_\omega, A(t) x_{id} + B(t) y_{id} \rangle$$
$$(1-t) A(t) \langle x_\omega, x_{id} \rangle + (1-t) B(t) \langle x_\omega, y_{id} \rangle + t A(t) \langle y_\omega, x_{id} \rangle + t B(t) \langle y_\omega, y_{id} \rangle$$

1. In Randers spaces, the distance from x to y is defined as the standard Riemannian distance plus a continuous line integral of a vector field W over the connecting curve γ
2. Given the great-circle angular distance between two identity features, we can define the spherical trajectory that is parameterized via SLERP
3. To evaluate the integral, we need the tangent velocity vector. Taking the direct analytical derivative of the SLERP formulation with respect to t yields:
4. The continuous ambient drift vector $W(\gamma(t))$ must be evaluated at every step. We approximate the intermediate drift field via linear interpolation of the anchor drift representations W_x and W_y
    1. Because these spatial interactions are decoupled from the time parameter t, they are pre-computed outside the integration loop via $O(1)$ batch matrix multiplications, drastically improving computational complexity.

[Distance 1] Using Slerp Interpolation

- **Spherical geodesic as path** : Taking spherical geodesic as the path between x and y
- **Field Assumption**: The drift vector field w changes linearly along this path

4

$$\omega(t) \approx (1-t)x_\omega + ty_\omega$$
$$\langle (1-t)x_\omega + ty_\omega, A(t)x_{id} + B(t)y_{id}\rangle$$
$$(1-t)A(t)\langle x_\omega, x_{id}\rangle + (1-t)B(t)\langle x_\omega, y_{id}\rangle + tA(t)\langle y_\omega, x_{id}\rangle + tB(t)\langle y_\omega, y_{id}\rangle$$

5

$$\mathcal{A}(\mathbf{x},\mathbf{y}) = \int_0^1 \langle \omega(\gamma(t)), \dot{\gamma}(t)\rangle dt \approx \sum_{k=1}^{K} \langle \omega(t_k), \dot{\gamma}(t_k)\rangle \Delta t$$
$$\Delta t = \frac{1}{K} \qquad t_k \in \{\frac{\Delta t}{2}, \frac{3\Delta t}{2}, ..., 1 - \frac{\Delta t}{2}\} \qquad K = 10$$

Critique: By defining the intermediate drift vector as such, we are linearly interpolating tangent vectors in the ambient Euclidean space. Mathematically inconsistent without applying Levi-Civita parallel transport

Employs a midpoint Riemann sum to discretely integrate the continuous path over K=10 steps

When numerically approximating the continuous integral over the interval $t \in [0, 1]$,the interval is discretized into K sub-intervals (bins), each with a width of $\Delta t = \frac{1}{K}$.

- Evaluating the continuous function at the exact center (midpoint) of each bin provides a significantly more accurate approximation than evaluating it at the left or right edges.

- **Bin 1:** Range $[0, \Delta t]$ .The midpoint is $0 + \frac{\Delta t}{2} = \frac{\Delta t}{2}$.
- **Bin 2:** Range $[\Delta, t2\Delta t]$ .The midpoint is $\Delta t + \frac{\Delta t}{2} = \frac{3\Delta t}{2}$.
- **Bin 3:** Range $[2, \Delta t3\Delta t]$ .The midpoint is $2\Delta t + \frac{\Delta t}{2} = \frac{5\Delta t}{2}$.
- ...
- **Bin K:** Range $[1, -\Delta t1]$ .The midpoint is $(1 - \Delta t) + \frac{\Delta t}{2} = 1 - \frac{\Delta t}{2}$.

By using the midpoint rule, the local error of the estimation is bounded by $O(\Delta t^2)$,effectively canceling out first-order linear errors compared to

standard left-endpoint or right-endpoint evaluations.

Critique:

- Tangent spaces at different points on a Riemannian manifold are disjoint vector spaces $T_x S^{n-1} \neq T_y S^{n-1}$
- You cannot linearly mix vectors from distinct tangent spaces without applying Levi-Civita parallel transport along the geodesic connecting them [Absil et al., Optimization on Matrix Manifolds 2008].

[Distance 1] Analytically solving the interpolation

Problem: Interpolate tangent vectors $x_\omega \in T_x\mathbb{S}^{n-1}$ and $y_\omega \in T_y\mathbb{S}^{n-1}$ along $\gamma(t)$

1 $T_{\gamma(t)}\mathbb{S}^{n-1}$, $\omega(t) = (1-t)P_{x\to\gamma(t)}(x_\omega) + tP_{y\to\gamma(t)}(y_\omega)$

2 $$\langle P(U), P(V)\rangle = \langle U, V\rangle$$

SLERP velocity $\dot\gamma(t)$ is the tangent vector field of $\gamma(t)$ -> it is parallel to itself $\nabla_{\dot\gamma}\dot\gamma = 0$

$\dot\gamma(t)$ Is the parallel transport of $\dot\gamma(0)$

3 $\langle P_{x\to\gamma(t)}(x_\omega), \dot\gamma(t)\rangle \equiv \langle x_\omega, \dot\gamma(0)\rangle$ $\langle P_{y\to\gamma(t)}(y_\omega), \dot\gamma(t)\rangle \equiv \langle y_\omega, \dot\gamma(1)\rangle$

$$\int_0^1 \langle \omega(t), \dot\gamma(t)\rangle dt = \int_0^1 \left[(1-t)\langle x_\omega, \dot\gamma(0)\rangle + t\langle y_\omega, \dot\gamma(1)\rangle\right] dt = \frac{1}{2}\langle x_\omega, \dot\gamma(0)\rangle + \frac{1}{2}\langle y_\omega, \dot\gamma(1)\rangle$$

1. To linearly interpolate tangent vectors along the geodesic gamma, we need to parallel transport both vectors to the intermediate tangent space, where $P_{A \to B}$ denotes the Levi-Civita parallel transport operator.
2. By the metric compatibility of the Levi-Civita connection, parallel transport preserves inner products
3. Because the SLERP velocity $\dot\gamma(t)$ is the tangent vector field of an affinely parameterized geodesic, it is parallel to itself $\nabla_{\dot\gamma}\dot\gamma = 0$ .Consequently, $\dot\gamma(t)$ is exactly the parallel transport of its initial velocity $\dot\gamma(0)$
4. The inner product with the transported drift vector simplifies to its boundary projection, which simplifies the integral of the geodesic to a linear function
5. Given the exact boundary derivatives of SLERP,

[Distance 1] Analytically solving the interpolation

3 $$\langle P_{x\to\gamma(t)}(x_\omega), \dot\gamma(t)\rangle \equiv \langle x_\omega, \dot\gamma(0)\rangle \qquad \langle P_{y\to\gamma(t)}(y_\omega), \dot\gamma(t)\rangle \equiv \langle y_\omega, \dot\gamma(1)\rangle$$

$$\int_0^1 \langle \omega(t), \dot\gamma(t)\rangle dt = \int_0^1 \left[(1-t)\langle x_\omega, \dot\gamma(0)\rangle + t\langle y_\omega, \dot\gamma(1)\rangle\right] dt = \frac{1}{2}\langle x_\omega, \dot\gamma(0)\rangle + \frac{1}{2}\langle y_\omega, \dot\gamma(1)\rangle$$

4 $$\dot\gamma(0) = \frac{\theta}{\sin\theta}(y_{\text{id}} - x_{\text{id}}\cos\theta) \qquad \dot\gamma(1) = \frac{\theta}{\sin\theta}(y_{\text{id}}\cos\theta - x_{\text{id}})$$

$$\langle x_\omega, x_{\text{id}}\rangle = 0 \qquad \langle y_\omega, y_{\text{id}}\rangle = 0$$

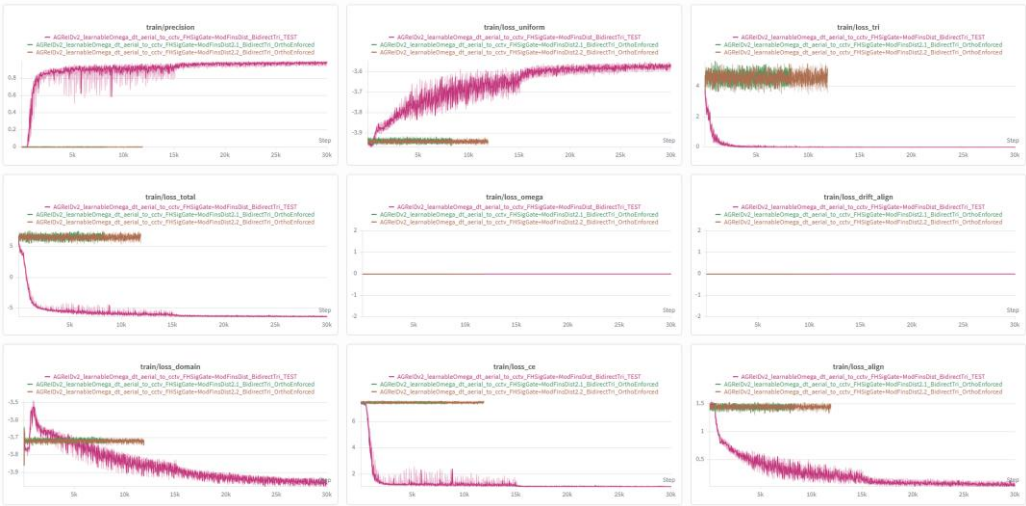
$$-\langle x_\omega, x_{\text{id}}\cos\theta\rangle = 0 \qquad \langle y_\omega, y_{\text{id}}\cos\theta\rangle = 0$$

5 $$\int_0^1 \langle \omega(t), \dot\gamma(t)\rangle dt = \frac{1}{2}\frac{\theta}{\sin\theta}\Big(\langle x_\omega, y_{\text{id}}\rangle - \langle y_\omega, x_{\text{id}}\rangle\Big)$$

Given the exact boundary derivatives of SLERP, we can assume strictly enforced tangent orthogonality to see that two terms evaluate to zero

The exact continuous integral resolves to

Results



- Loss curves are flatlining at exactly 0 and invariant total/triplet/CE losses indicate gradient explosion leading to network collapse

Hypothesis 1

The formulation introduces two well-documented numerical singularities in Riemannian backpropagation [Nickel & Kiela, NeurIPS 2017]:

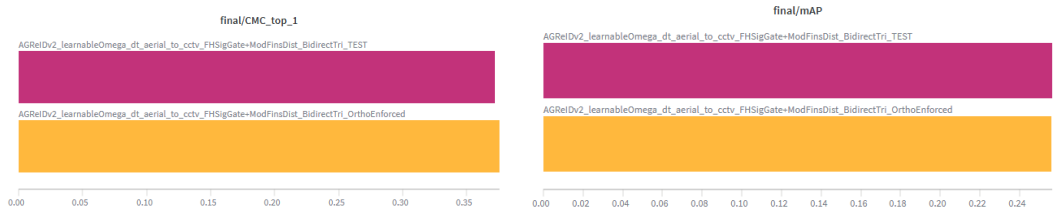
- **The Arccosine Singularity:** torch.acos(cos_theta) has a derivative of $-1/\sqrt{1-x^2}$
    - As cos_theta approaches 1, the gradient approaches infinity. Clamping to 1.0 - 1e-6 yields a maximum gradient multiplier of ~ 707
    - In a Triplet Loss formulation, this massive scalar propagates backward, causing an immediate weight explosion.
- **The Indeterminate Ratio:** The term theta / sin_theta evaluates to $0/0$ as $\theta \to 0$. Even with clamping sin_theta to 1e-6, if theta evaluates to 0.0, the gradient calculation inside PyTorch's auto-grad graph becomes highly unstable.

Detach the arccos?

[Model] Enforcing Tangent Plane Orthogonality

- For a point $x \in S^{n-1}$, the tangent space is $\{v \in R^n \mid \langle x, v \rangle = 0\}$
- **Apply** orthogonal projection operator of the hypersphere

The exact projection onto the tangent plane $T_x S^{n-1}$ is defined as $x_\omega = (I - x_{\mathrm{id}} x_{\mathrm{id}}^T)\hat{x_\omega}$



For a point $x \in S^{n-1}$, the tangent space is $\{v \in R^n \mid \langle x, v \rangle = 0\}$.To strictly enforce this, you must apply the orthogonal projection operator of the hypersphere[Absil et al., Optimization on Matrix Manifolds 2008].

Did not yield big results

[Regularization] Preventing Omega from Diverging

We use the logarithmic barrier penalty

$$s_i = \max(\epsilon, c - \|\omega_i\|_2)$$

Where c is the maximum norm defined (0.95).

$$\mathcal{L}_{\text{barrier}} = -\frac{1}{B}\sum_{i=1}^{B} \log(\max(\epsilon, c - \|\omega_i\|_2))$$

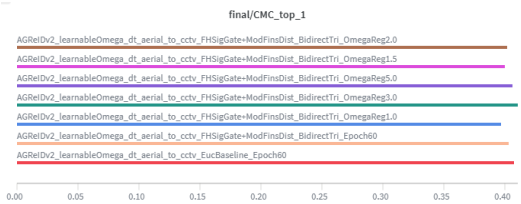
$$s_i \to 0 \qquad -\log(s_i) \to \infty$$

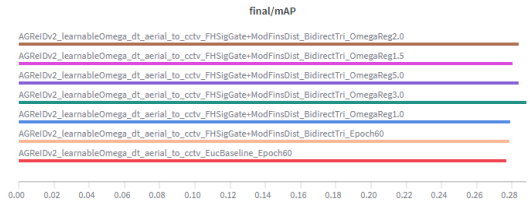
Gradients of this loss w.r.t. norm is:

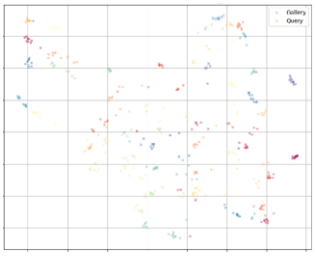
$$\frac{\partial \mathcal{L}_{\text{barrier}}}{\partial \|\omega_i\|_2} = \frac{1}{c - \|\omega_i\|_2}$$

- The slack variable s_i represents the distance from the norm to the boundary
    - To prevent numerical instability when evaluating the logarithm at exactly zero, a clamping parameter ϵ (eps = 1e-6) is introduced
- The final empirical loss across the batch is the mean of the negative logarithms of these slack variables
- From an optimization perspective, as the norm $\| \omega_i \|_2$ approaches the boundary c, the slack $s_i \to 0$,and the term $-\log(s_i) \to \infty$.
- This indicates that the restorative force repelling the vector from the boundary grows asymptotically to infinity as the vector approaches the boundary, acting as a strict "hard constraint."

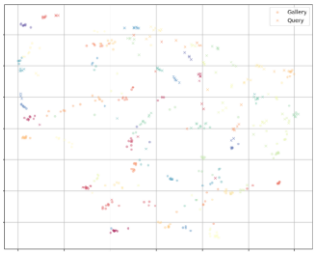
[Regularization] Preventing Omega from Diverging







OmegaReg=3.0



Baseline

[**Theory**]: Explain why drift align and bidirectional triplet are incompatible

- Applying both constraints simultaneously overconstrains the latent geometry
- Model does not have enough “slack” to model asymmetry?

Not quite sure why

[**Theory**]: Explain why having a 32-dimensional drift vector is optimal

Found that the premise of integrating an asymmetric Finsler space into a domain-generalizable person Re-ID space is theoretically flawed?

- Person Re-ID is fundamentally an identity retrieval task, which necessitates a equivalence relation governed by metric symmetry
- Introducing distance asymmetry inherently violates identity equivariance, as the retrieval ranking becomes strictly dependent on the query-gallery orientation.

Drift vector of size 32 is "optimal" and surpasses the Euclidean baseline:

- not indicative of the model successfully learning meaningful domain-based asymmetry
- an artifact of information bottlenecking and implicit feature-space regularization

1. Capacity Constraint on Asymmetry
2. Topological Perturbation vs. Uniformity Collapse

Capacity Constraint on Asymmetry

- asymmetry fundamentally degrades the symmetric identity matching objective
- slicing the drift vector to a restricted dimensional subspace (e.g., 32 dimensions) heavily limits the destructive impact of the asymmetric term in the Randers metric
- Bounding the drift to 32 dimensions forces the network to allocate the vast majority of the latent space capacity to the symmetric Riemannian (Euclidean) metric, effectively preserving the identity equivalence class.
- Increasing the drift dimension (e.g., 64, 128, 2048) allocates excessive capacity to the directional bias, catastrophically degrading symmetric retrieval performance as demonstrated in your mAP and Rank-1 drops.

Topological Perturbation vs. Uniformity Collapse:

- The BAU framework mitigates the polarized effects of data augmentations by explicitly balancing alignment and uniformity on an isotropic unit hypersphere.
- The Randers metric introduces a local preferred direction via the drift vector, which inherently breaks this required hyperspherical isotropy.
- A high-dimensional drift (e.g., ≥ 64) aggressively warps the manifold

    - Encouraging spatial sparsity and representation collapse that the BAU uniformity loss cannot recover from.
- A 32-dimensional bottleneck acts as an optimal regularizer
    - it provides enough geometric perturbation to prevent source-domain overfitting
    - while maintaining sufficient global isotropy for the uniformity loss to converge.
    - A 16-dimensional drift simply lacks the parameterization to exert meaningful regularization gradients over the Euclidean baseline.

This Week

1. [Literature] Paper presentation: ReNorm
2. [Literature] Paper presentation: CLIP-FDGI
3. [Sweep] Omega Regularization Sweep
4. [Model] Attempts at fixing the SLERP / analytical distance computation
5. [Test] Benchmarked the Finsler BAU against other papers in the same field
6. [Model] Domain-Conditioned Global Drift

Tabled for next week

1. [Model] Devise a plan to migrate Finsler modifications to CLIP-FDGI / ReNorm
2. [Visualisation] Plot the omega drift vector
3. [Model] Manifold Smoothness Regularization: Force the network's vector field to match the surrogate.
   - During training, sample the midpoint of the geodesic $\gamma(0.5)$
   - Pass this midpoint through the neural network to obtain the true intermediate drift vector ω_{true}
   - Compute the parallel-transported surrogate vector at the midpoint, $\omega_{surrogate}$.
   - Add a regularization loss: $L_{smooth} = \| \omega_{true} - \omega_{surrogate} \|_2^2$

[Literature] ReNorm

ReNorm: Re-evaluating Instance Normalization for DG Re-ID

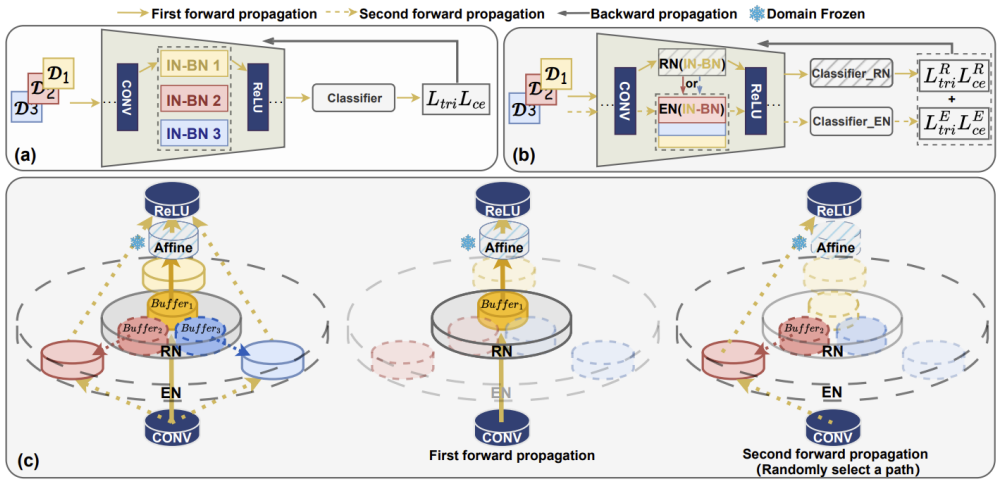
- **The Problem:** Standard Instance Normalization (IN) indiscriminately flattens feature variances, inadvertently destroying fine-grained identity structures (e.g., clothing textures) alongside domain styles.
- **The Solution:** Introduces an adaptive, channel-wise regularization mechanism to modulate variance reduction.
- **Mechanism:** Identifies and mathematically protects variances highly correlated with identity discrimination, while standardizing only those strictly tied to domain style.

Comparing with BAU:

- **Strategy:** ReNorm operates at the *architectural feature-map* level (manipulating mean/variance statistics prior to pooling), whereas BAU operates entirely at the *output embedding metric* level.
- **Philosophy:** ReNorm believes domain gaps are best handled by selectively removing style at the CNN/Transformer intermediate layers. BAU believes domain gaps are best handled by repelling domain clusters directly on the unit hypersphere.

- *Context:* In typical Domain Generalization for Re-ID, networks try to filter out domain-specific noise.
    - Methods like IBN-Net use standard normal distributions $N(0, 1)$ to achieve this.
- *Elaboration:* ReNorm shows that flattening σ^2 is destructive.
    - It proposes an attention-like parameter λ_c to selectively scale the normalized feature, retaining the residual identity signal.
- *Connection to our work (BAU/Finsler):*
    - ReNorm highlights a critical flaw in traditional DG Re-ID: treating domain bias purely as noise to be erased harms retrieval.
    - Our Finsler approach takes the opposite, constructive route
        - we explicitly isolate the domain bias into an orthogonal drift slice x_{drift} rather than attempting a destructive normal mapping.

[Literature] ReNorm



Directing Attention (The Standard Approach):
top left shows how traditional domain-generalizable Re-ID networks handle feature normalization.

- They typically interleave Batch Normalization (BN) and Instance Normalization (IN) layers.
- As the text snippet confirms, while BN helps with discriminative features, it overfits to source domains. IN is used to eliminate that domain-specific style.

The Critical Flaw:
The core problem that ReNorm highlights is:

- treating domain bias purely as noise to be erased harms retrieval.
- When standard IN standardizes the feature variances to remove 'domain style',
- it wipes out the fine-grained, discriminative identity structures—like clothing textures—that are crucial for distinguishing between different people.

The ReNorm Solution (The Dual Path):
To solve this, ReNorm steps away from static IN-BN blocks.

- As shown in the right and bottom graphics, it uses a unique dual forward-

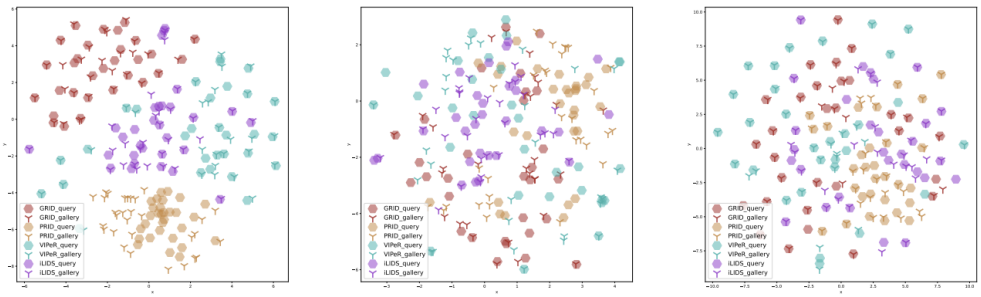
propagation strategy with the exact same network weights.

- In the first pass, it uses **RN (Representation Normalization)** to focus purely on extracting the most highly discriminative identity features within the source domains.
- In the second pass, it uses **EN (Exploration Normalization)** to actively force those weights to generalize to unseen target domains."

Our Narrative Takeaway:
ReNorm illustrates that destroying domain information at the feature level is mathematically highly destructive to identity tracking.

- While ReNorm attempts to fix this by balancing two complex normalization paths, our repository takes a cleaner geometric approach:
- we don't try to destroy the domain bias at all. Instead, we explicitly isolate it into a dedicated Finsler drift vector, preserving 100% of the identity branch's structural variance."

[Literature] ReNorm



(a) BN (b) IN (c) ReNorm (ours)

- "BN" model exhibits smaller distances between query and gallery sets of the same identity but is heavily influenced by domain factors, leading to clear domain boundaries.
- Fig.4 (b) shows that IN removes these boundaries but increases the distance between query and gallery sample pairs.

The observations prove that BN can capture more discriminative features but tends to overfit to source domains, while IN can eliminate domain-specific information to enhance generalization but inevitably removes discriminative information.

- Fig.4 (c) shows that the distances between query and gallery sample pairs are much smaller
- Simultaneously, there are no apparent boundaries among target domains, indicating that our method effectively eliminates domain-specific information.

[Literature] CLIP-FDGI

CLIP FDGI: Fine-grained Domain Generative Interchange

- **The Problem:** Purely visual domain gaps (e.g., lighting, extreme camera angles) remain difficult to bridge without computationally heavy pixel-level GANs or feature mixups.
- **The Solution:** Leverages Vision-Language Models (CLIP) to perform "Generative Interchange" in the semantic text space.
- **Mechanism:** Disentangles prompts into [ID Token, *Domain Token*]. By swapping the domain token while keeping the ID token constant, it synthesizes target-domain proxies constrained by an InfoNCE loss.

Comparing with BAU:

- **Strategy:** FDGI augments the training domain distribution virtually using text-driven prompt-swapping. BAU utilizes only the existing multi-source visual domains and explicitly forces distance between them.
- **Philosophy:** FDGI tackles unseen domains by trying to conceptually simulate them during training via the vast prior knowledge of CLIP. BAU tackles unseen domains by regularizing the space to be tightly alignment-focused, hoping the hypersphere uniformity generalizes naturally.

- *Context:* With the rise of foundational models, FDGI maps Re-ID features to contrastive text embeddings to bridge domains visually.
- *Elaboration:* Instead of faking images, FDGI forces the image encoder to disentangle identity from domain
    - by contrasting against synthetically swapped cross-modal prompts.
- *Connection to our work (BAU/Finsler):* FDGI validates our core hypothesis: identity and domain *must* be explicitly disentangled.
    - Just as FDGI uses a learned domain token in language space, our DomainConditionedDriftHead infers a soft domain prior in the visual metric space.
    - We achieve systematic domain disentanglement geometrically, without the extreme computational overhead of multi-modal architectures.

[Model] Omega Regularization Sweep

	mAP		Rank 1		Mean Norm	Max Norm
Euclidean	42.21		42.5			
0.5	41.62	-0.59	42.21	-0.29	0.2636	0.3105
0.75	42.81	+0.6	42.14	-0.36	0.1841	0.2283
1.0	43.07	+0.86	43.57	+1.07	0.1294	0.1589
1.5	43.95	**+1.74**	43.93	+1.43	0.0762	0.1001
2.0	43.43	+1.22	43.5	+1	0.0553	0.0677
2.5	43.69	+1.48	44.14	+1.64	0.0448	0.0531
3.0	43.7	+1.49	44.43	**+1.93**	0.0423	0.0468
5.0	42.23	+0.02	42.93	+0.43	0.0386	0.0439

Small omega norm / mean means either:

1. Asymmetry is not useful in current setup
2. Current optimization setup makes the drift branch too weak to matter

[Model] Attempts at Fixing SLERP

Root Cause: The Arccosine Singularity
Gradient cascades to infinity near identical or diametrically opposed embeddings

$$\frac{d}{dx}\arccos(x) = \frac{-1}{\sqrt{1-x^2}} \xrightarrow{x\to\pm 1} -\infty$$

Solution A: Bounded Linear Extrapolation for arccos
Instead of naïve clamping (results in uninformative zero-gradients at boundaries), switched to tangential linear extrapolation away from singularity.

$$\check{\arccos}(x) = \begin{cases} \arccos(B) - \frac{x-B}{\sqrt{1-B^2}} & \text{if } x > B \\ \arccos(x) & \text{if } |x| \le B \end{cases}$$

Adopted from SO(3) handling found in PyTorch3D (*acos_linear_extrapolation*)

Solution B: Small-Angle Taylor Expansions
Prevent 0/0 NaNs in the velocity overlap integral via smooth series expansion

$$\frac{\theta}{\sin\theta} \approx 1 + \frac{\theta^2}{6} + \frac{7\theta^4}{360} \quad \text{for } |\theta| < 10^{-3}$$

- When modeling our Finsler drift via SLERP or analytical integration, we noticed immediate model divergence.
    - culprit lies in slerp/analytical finsler distance: we require the angle θ, derived from $\arccos(\cos\theta)$
    - Because our identity features are unit-normalized, positive pairs naturally yield a cosine similarity extremely close to 1.
    - In backpropagation, the derivative of arccps features a $\sqrt{1-x^2}$ in the denominator.
    - As x approaches 1, our gradients explode to infinity. Under mixed-precision, this instantly generates NaNs.

- Previous fixes just clamped the inputs
    - But clamping sets gradients exactly to zero at the threshold, which causes stalled learning.
    - So, we looked to state-of-the-art geometry libraries like PyTorch3D.
    - We implemented a linearly extrapolated arccosine.
        - When the cosine similarity exceeds a strict safety bound B, we freeze the curvature and smoothly follow the tangent line
        - maintaining a stable, non-zero gradient all the way through.

    - B is 1.0 - 1e-4 if fp16, otherwise 1.0 - 1e-5
    - To doubly fortify this, these sensitive angular matrix overlaps are locally forced from FP16 back into strict FP32 precision to grant us the widest representational margin possible.

- Finally, our asymmetry integrals rely heavily on the ratio of θ over $\sin(\theta)$.
    - For extremely small angles—again, highly common during convergence—this creates an unstable $0/0$ division.
    - Instead of a hard, flat mask which ignores local curvature, we adopted operations standardized by Riemannian frameworks like Geoopt.
    - We swap out the direct calculation for its 4th-order Taylor Expansion below a set angular threshold limit.
    - This ensures the geometric traversal stays continuously smooth and computationally bounded, fully eliminating our divergence issues with training collapse during training

[Experiment] Protocol 2 Head to Head

	M+MS+CS → C3		M+CS+C3 → MS		MS+CS+C3 → M		Average	
	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
CLIP-FGDI	**44.4**	44.6	**31.1**	**59.4**	**79.4**	**91.3**	**51.6**	**65.1**
ReNorm	43.6	44.7	25.6	55.6	72.7	89.1	47.3	63.1
ReNorm*	42.4	41.5	24.2	53.3	72.6	88.9	46.4	61.2
BAU	42.8	43.9	24.3	50.9	77.1	90.4	48.1	61.7
BAU*	42.21	**49.70**	23.28	42.5	76.61	89.61	47.37	60.60
Finsler BAU*	43.95	48.04	22.22	43.93	76.47	89.91	47.55	60.63

Sample-level Finsler asymmetry is not strongly supported in DG person ReID; however, domain-structured directional bias remains meaningful and should be modeled as a nuisance factor separated from identity. To continue, we need to consider the following angles:

- *When does asymmetry help*?
- *At what level should it be modeled*?
- *Why does DG training collapse it towards near-Euclidean behavior*?

- Finsler BAU is effectively tied with BAU on the average -> too small to support a strong "asymmetric geometry clearly improves DG person ReID" claim

- Training intentionally suppresses instability in drift branch: reduced drift LR, clipping, omega regularization and removing the drift vector from the alignment loss
- While DG objective itself wants identity invariance -> conflicts with an identity-based directional drift signal

1. When does asymmetry help?

Asymmetric distances (such as Randers/Finsler metrics) are advantageous strictly when the underlying information matching is non-commutative.

- In person Re-ID, metric asymmetry is beneficial when moving from an information-poor domain (e.g., low-resolution aerial probes) to an information-rich domain (e.g., high-resolution ground CCTV).
- It compensates for the fact that feature containment is directional: the features visible in an aerial shot are a subset of those in a close-up, but the reverse is not true.

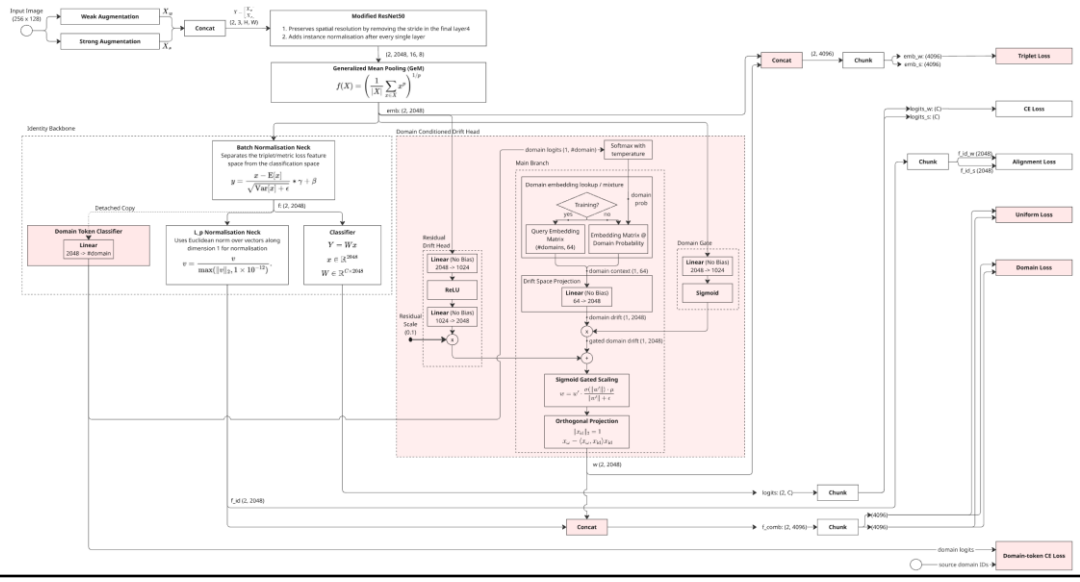
2. At what level should it be modeled?
Asymmetry should be modeled at the **domain, view, or modality level**, not the instance level.

- Formulating a Finsler drift vector strictly on a per-identity or per-image basis parameterizes instance-specific noise rather than systematic geometric bias.
- Literature on disentangled representations indicates that nuisance factors (like camera angle, illumination, or sensor modality) are shared across sets of images.
    - *Kulis, B., et al. "Asymmetric metric learning." ICML, 201*
- Thus, the asymmetric drift vector must be decoupled from the identity representation and conditioned on the domain or camera view to act as a proper parallel-transport mechanism across the manifold.

3. Why does DG training collapse it towards near-Euclidean behavior?

- Domain Generalization (DG) objectives—specifically alignment and uniformity constraints (such as those in BAU)—function by explicitly penalizing domain-specific structure to enforce a domain-invariant latent space.
- If an asymmetric drift vector captures domain bias, the DG domain-invariance regularizers will view this drift as a nuisance feature and optimize it toward zero.
- Furthermore, deep neural networks exhibit "simplicity bias" under stochastic gradient descent; if a symmetric Euclidean distance is sufficient to separate identities, the network will discard the more complex Finsler gradient pathways, resulting in a metric collapse to the Euclidean baseline.

[Model] Architecture Proposal: Domain Conditioned Global Drift



- Previously, our *Finsler Drift Head* generated an asymmetric drift vector for every single instance independently.
- Our experiments revealed that Domain Generalization (DG) objectives naturally suppress this.
    - Networks exhibit a "simplicity bias" (*Shah et al., 2020*); if a symmetric metric works well enough, the network zeros out the complex instance-level drift.
- In real deployments (e.g., aerial vs. CCTV), asymmetry is a *domain/view-level property*, not an identity property.
- Therefore, we've redesigned the drift head to explicitly condition the asymmetric geometry on the domain/view structure.

The Foundation: The Identity Pathway

(Point to the top/left of the diagram)

- We maintain our modified ResNet50 (removing the final stride) and GeM pooling (*Radenović et al., 2018*) to preserve fine-grained spatial cues.
- The Batch Normalization neck and L_2 normalization cleanly project the identity into a view-invariant unit hypersphere.
    - **Role:** This branch acts as our rock. It focuses *only* on identity

discrimination.

The Core Innovation: Domain Prediction & Soft Mixtures
Point to the "Domain Token Classifier" (Pink box, left).

- We branched off a *detached* copy of the identity feature into a linear Domain Classifier. This prevents domain-supervision gradients from corrupting the core identity backbone (*Ganin & Lempitsky, 2015*).

Point to "Domain embedding lookup" (Center).

- During training, we know the source domains.
- But at evaluation on an *unseen* target domain, this classifier uses a temperature-scaled softmax to infer a soft probability mixture
    - (e.g., "This looks 70% CCTV, 30% Wearable").
- Instead of generating drift from scratch, we use those probabilities to mix learned domain embeddings.
- This enforces a crucial inductive bias: asymmetric drift is *shared* across identities based on view.

Adaptive Modulation: Gating and Residuals
Point to the "Domain Gate" and "Residual Drift Head".

- Not every image needs full domain drift.
    - A small neural gate dynamically scales the domain context so the model can gracefully fall back on identity for atypical samples (*Hu et al., 2018*).
- We still allow a tiny, per-instance residual correction (scaled down to 0.1) for lighting/pose noise.
    - This keeps the domain structure dominant while providing necessary flexibility.

Strict Disentanglement: Post-Processing
Point to the bottom of the Drift Head (Sigmoid Scaling & Orthogonal Projection).

- The sigmoid-gated scaling controls the total magnitude of the drift vector globally, allowing the model to shrink asymmetry to zero if it doesn't aid a specific batch.
- With the orthogonal projection, we mathematically force the drift vector to be orthogonal specifically to the identity vector
    - It ensures drift is strictly a nuisance/view factor.
    - It absolutely prevents the model from using drift as a shortcut to encode identity (*Kumar et al., 2018*).

Loss Integration & Conclusion
Point to the right side (Loss functions).

- BAU's Alignment loss strictly govern the identity slice to keep it augmentation invariant.
- Domain and Uniformity loss makes the embedding domain invariant.
- Our new Domain-Token CE loss trains the domain predictor.

Asymmetric Triplet: Finally, the full concatenated [Identity + Drift] vector is fed into the Triplet Loss. Here, our asymmetric Finsler distance performs direction-aware hard-negative mining.

Conclusion: By explicitly factoring the embedding into a domain-invariant identity and a domain-conditioned directional drift,

- this architecture solves the instance-collapse problem.
- It allows us to apply structured, generalizable asymmetry to completely unseen camera domains.

Negative samples should come from different domains

- Same identity from different domains using finsler distances
- Inject intuition from the equation
- Start simpler – add constrastive loss terms, then change the model

[Model] Architecture Proposal: Domain Conditioned Global Drift

Why the scaled residual?
Pure domain conditioning assumes every camera angle within a domain is perfectly identical.
- scaled residual (λ=0.1) allows the network to make micro-adjustments for localized nuisances (lighting, specific occlusions)
- while enforcing a strict mathematical bias that the overarching domain structure governs the primary asymmetric geometry
  - prevent the model from collapsing back to instance-level overfitting.

Why does orthogonality prevents shortcut learning?
- Neural networks suffer from "Simplicity Bias" (*Geirhos et al., 2020*)
- With the orthogonal projection, the drift vector is stripped of all identity variance
- Drift branch has to model whatever variance is left to minimize the loss on combined embedding

The Threat of the "Lazy" Network: Neural networks suffer from "Simplicity Bias" (*Geirhos et al., 2020*). They achieve the lowest loss using the easiest features available.
- Because the identity vector (x_{id} (is heavily supervised to perfectly separate individuals, it is the most powerful signal in the network.
- If we just concatenate the drift vector (ω) directly, the easiest way for the network to satisfy the Triplet Loss is to lazily copy the identity features into the drift branch.

We explicitly subtract the projection of ω onto x_{id} :$\omega' = \omega - \langle \omega, x_{id} \rangle x_{id}$.
- The dot product is zero. The drift vector is mathematically stripped of *all* identity variance.
- To minimize the loss on the combined embedding now, the drift branch *has no choice* but to model whatever variance is left—which translates perfectly to camera view, angle, and perspective asymmetry.

SOTA Grounding:

Domain Separation Networks: Pioneered by *Bousmalis et al., 2016, NeurIPS*. They proved that without explicit "Difference Losses"

(orthogonality), private domain branches collapse and simply duplicate shared identity features.

Re-ID Disentanglement: Standard practice to prevent metric corruption when separating nuisance factors (like pose) from core identity (*Karanam et al., 2019, IEEE TPAMI*).

[Literature] Orthogonal Projection Hypersphere

Example 3.6.1 ***Sphere***

On the unit sphere S^{n-1} considered a Riemannian submanifold of $\mathbb{R}^n$, the inner product inherited from the standard inner product on $\mathbb{R}^n$ is given by

$$\langle \xi, \eta \rangle_x := \xi^T \eta. \tag{3.33}$$

The normal space is

$$(T_x S^{n-1})^\perp = \{x\alpha : \alpha \in \mathbb{R}\},$$

and the projections are given by

$$\mathrm{P}_x \xi = (I - xx^T)\xi, \qquad \mathrm{P}_x^\perp \xi = xx^T \xi$$

for $x \in S^{n-1}$.

```python
def proju(self, x: torch.Tensor, u: torch.Tensor) -> torch.Tensor:
    u = u - (x * u).sum(dim=-1, keepdim=True) * x
    return self._project_on_subspace(u)
```

Next Week

1. [Model] Devise a plan to migrate Finsler modifications to CLIP-FDGI / ReNorm
2. [Model] Experiment with the DomainConditionedFinslerHead
3. [Visualisation] Plot the omega drift vector
4. [Model] Manifold Smoothness Regularization: Force the network's vector field to match the surrogate.
   - During training, sample the midpoint of the geodesic $\gamma(0.5)$
   - Pass this midpoint through the neural network to obtain the true intermediate drift vector ω_{true}
   - Compute the parallel-transported surrogate vector at the midpoint, $\omega_{surrogate}$.
   - Add a regularization loss: $L_{smooth} = \| \omega_{true} - \omega_{surrogate} \|_2^2$